Possible Problems for Quiz 4 to be taken 2/7/19. Of these problems I will choose 3 to be on the quiz.

Problem 1

Multiply: $\begin{bmatrix} 1 & 2 & 0 \\ -1 & 0 & 3 \\ 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 2 & -1 \end{bmatrix}$

Problem 2

Prove $(A+B)^T = A^T + B^T$

Problem 3

Consider the linear system $2x_1 - 3x_2 = 1$ and $x_1 - x_2 = 4$ Write the system in the form $A\vec{x} = b$ and find the solution using A^{-1}.

Problem 4

For the matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ what condition can be placed on a, b, c, d which will guarantee that A is **not** invertible? Justify your answer.

Problem 5

Let $A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix}$ find $AAAA$. What is A^{-1}?

Problem 6

a) Suppose E_1, E_2, E_3 are elementary matrices such that $E_3E_2E_1A$ puts the **square matrix** A into reduced row echelon form. What is one way you could write the product $E_3E_2E_1$ in terms of A? Justify your answer.

b) Suppose A, B, C are all invertible matrices of the same size. What is $(ABC)^{-1}$?Justify your answer.